```python
import re
from collections import Counter

# Customer reviews dataset
reviews = [
    "The product is excellent and very useful",
    "I love this product, it is excellent",
    "The product is good and easy to use",
    "Very useful product and good quality",
    "I am happy with this excellent product"
]

# Combine all reviews
text = " ".join(reviews)

# Convert to lowercase
text = text.lower()

# Remove punctuation
text = re.sub(r'[^\w\s]', '', text)
# Split into words
words = text.split()

# Calculate word frequency
frequency = Counter(words)

# Display frequency distribution
print("Word Frequency Distribution:")
for word, count in frequency.items():
    print(word, ":", count)
```

```
Word Frequency Distribution:
the : 2
product : 5
is : 3
excellent : 3
and : 3
very : 2
useful : 2
i : 2
love : 1
this : 2
it : 1
good : 2
easy : 1
to : 1
use : 1
quality : 1
am : 1
happy : 1
with : 1
```

```python
import pandas as pd
import re
import matplotlib.pyplot as plt
from collections import Counter
from google.colab import files

# ---------------------------------------
# 1. Upload CSV Dataset
# ---------------------------------------

print("Upload your CSV file:")

uploaded = files.upload()

file_name = list(uploaded.keys())[0]

print("\nFile uploaded successfully:", file_name)

# ---------------------------------------
# 2. Load Dataset
# ---------------------------------------

df = pd.read_csv(file_name)

print("\nDataset loaded successfully!")
print("Number of records:", len(df))

print("\nDataset Preview:")
print(df.head())
```

```python
# --------------------------------------
# 3. Check feedback column
# --------------------------------------

if "feedback" not in df.columns:
    print("\nERROR: CSV must contain a column named 'feedback'.")

else:

    # --------------------------------------
    # 4. Stop Words
    # --------------------------------------

    stop_words = {
        "the", "and", "is", "a", "an", "to", "of", "in",
        "for", "on", "with", "this", "that", "it", "was",
        "are", "as", "at", "be", "by", "from", "or", "but",
        "not", "very", "i", "you", "we", "they", "he",
        "she", "my", "our", "your", "me", "so", "have",
        "has", "had"
    }

    # --------------------------------------
    # 5. Preprocess Feedback
    # --------------------------------------

    words = []

    for text in df["feedback"].dropna():

        # Convert to lowercase
        text = text.lower()

        # Remove punctuation and numbers
        text = re.sub(r"[^a-zA-Z\s]", "", text)

        # Split into words
        text_words = text.split()

        # Remove stop words
        for word in text_words:
            if word not in stop_words:
                words.append(word)

    # --------------------------------------
    # 6. Calculate Word Frequency
    # --------------------------------------

    frequency = Counter(words)

    # --------------------------------------
    # 7. Get N from User
    # --------------------------------------

    while True:

        try:
            N = int(input("\nEnter the number of top words (N): "))

            if N <= 0:
                print("Please enter a positive number.")
            else:
                break

        except ValueError:
            print("Invalid input!")
            print("Please enter only a number, for example: 5, 10, or 15.")

    # --------------------------------------
    # 8. Get Top N Words
    # --------------------------------------

    top_words = frequency.most_common(N)

    # --------------------------------------
    # 9. Display Results
    # --------------------------------------

    print("\nTop", N, "Most Frequent Words")
    print("--------------------------------")

    for word, count in top_words:
        print(word, ":", count)
```

```
    print(word, ":", count)

# ---------------------------------------
# 10. Plot Bar Graph
# ---------------------------------------

words_list = [word for word, count in top_words]
counts = [count for word, count in top_words]

plt.figure(figsize=(10, 6))

plt.bar(words_list, counts)

plt.xlabel("Words")
plt.ylabel("Frequency")

plt.title("Top N Most Frequent Words in Customer Feedback")

plt.xticks(rotation=45)

plt.tight_layout()

plt.show()
```

```
Upload your CSV file:
Choose Files data.csv
```

```
Start coding or generate with AI.
```

```
File uploaded successfully: data.csv

Dataset loaded successfully!
Number of records: 50

Dataset Preview:
                                           feedback
0          The product is excellent and easy to use
1  I love this product and the quality is excellent
2    The product is very useful and works perfectly
3           Good product with excellent performance
4                 I am very happy with this product

Enter the number of top words (N): 10

Top 10 Most Frequent Words
--------------------------------
product : 33
excellent : 23
good : 19
quality : 15
easy : 10
use : 10
```